

Project Report: MLOps-Driven Doctor Recommendation System – CareCompass

1. Introduction

CareCompass is an end-to-end **Machine Learning Operations (MLOps)** enabled project that automates the development, deployment, and monitoring of a **Doctor Recommendation System**.

The goal is to help patients receive the best-matched doctor suggestions based on symptoms, specialty requirements, patient history, and predictive analytics — all while ensuring continuous integration, delivery, and automated scaling using Kubernetes.

2. Project Overview

The system integrates:

- **Machine Learning (Recommendation Model)**
- **Backend Application (Flask & FastAPI)**
- **CI/CD Pipeline (Jenkins + GitHub)**
- **Containerization (Docker)**
- **Orchestration (Minikube)**
- **Infrastructure as Code & Automation (Ansible)**

It follows modern MLOps principles such as automated training, reproducible builds, version-controlled models, and seamless deployment.

3. Problem Statement

Patients often struggle to identify the right doctor based on:

- Symptoms
- Disease category
- Specialty needed
- Location constraints
- Historical success rates

Traditional search-based systems lack personalization.

CareCompass solves this with **ML-driven recommendations** deployed through an automated MLOps pipeline.

4. Architecture Overview

4.1 End-to-End Workflow

1. **Developer pushes code to GitHub**
 2. Jenkins automatically:
 - pulls repo
 - installs dependencies
 - trains/validates model
 - builds Docker image
 - pushes to Docker Hub
 - triggers Ansible or Kubernetes deployment
 3. User requests prediction → service returns **doctor recommendations**
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5. Machine Learning Component

5.1 Model Purpose

The ML model predicts the **top recommended doctors** based on input symptoms or condition.

5.2 Features Used

Patient/Disease Features(Disease-to-Specialty Mapping)

- disease
- symptoms
- disease description (reason)
- tests/procedures
- medications
- total severity score (sometimes used)

Doctor Features(Ranking Doctors within that Specialty)

- specialty
- doctor rating
- experience
- distance from patient
- availability
- cost of services

These features together power the recommendation engine. Using features from `updated_dataframe.csv`, the system predicts **which doctor specialty** is best for the patient's condition. Using features from `doctors_data_multireward.csv`, doctors are ranked.

5.3 Model Type

Typical formulations include:

- **Classifier** to predict specialty
- **Similarity-based recommender** (cosine similarity / embeddings)
- **Ranking model** for best doctor match

5.4 Output

The system returns:

- Doctor name
- Specialty
- Estimated confidence score(Rating)

6. Use Cases

1. **Online health consultation apps**
 2. **Symptom-based triaging and routing**
 3. **Personalized patient experience**
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7. Conclusion

CareCompass represents a complete MLOps-driven solution that merges machine learning, automation, containerization, and scalable deployment.

It showcases how modern engineering practices can turn an ML model into a reliable, continuously improving production-grade recommendation system.

